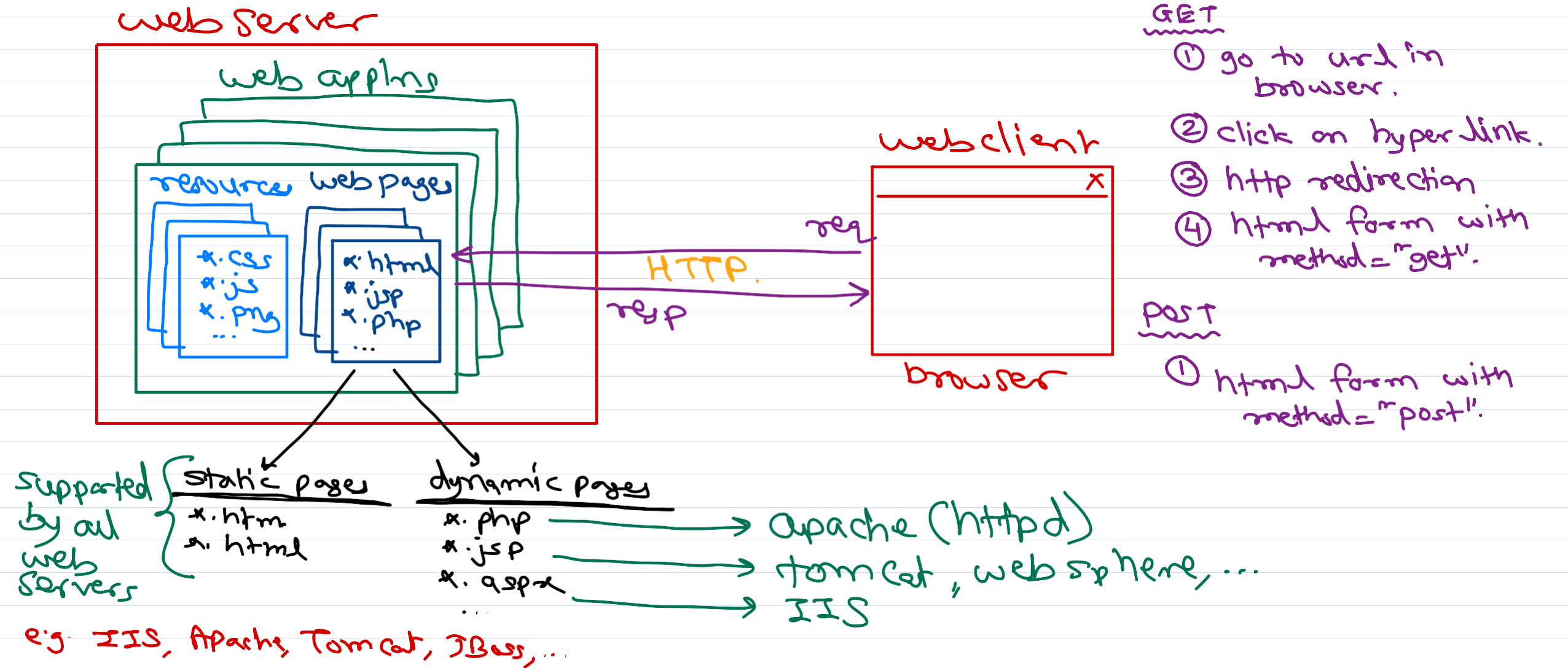
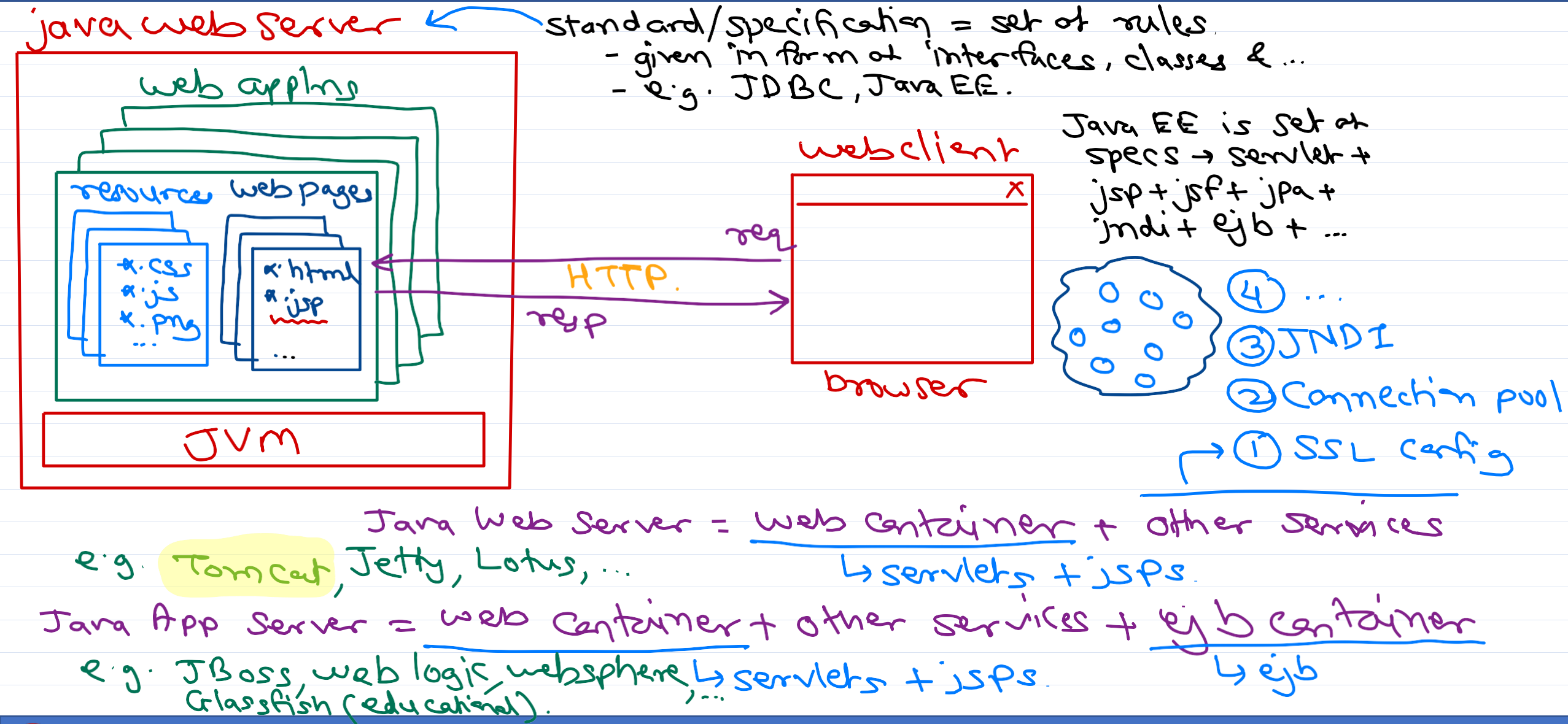


Web Server



Java Web Server



Apache Tomcat

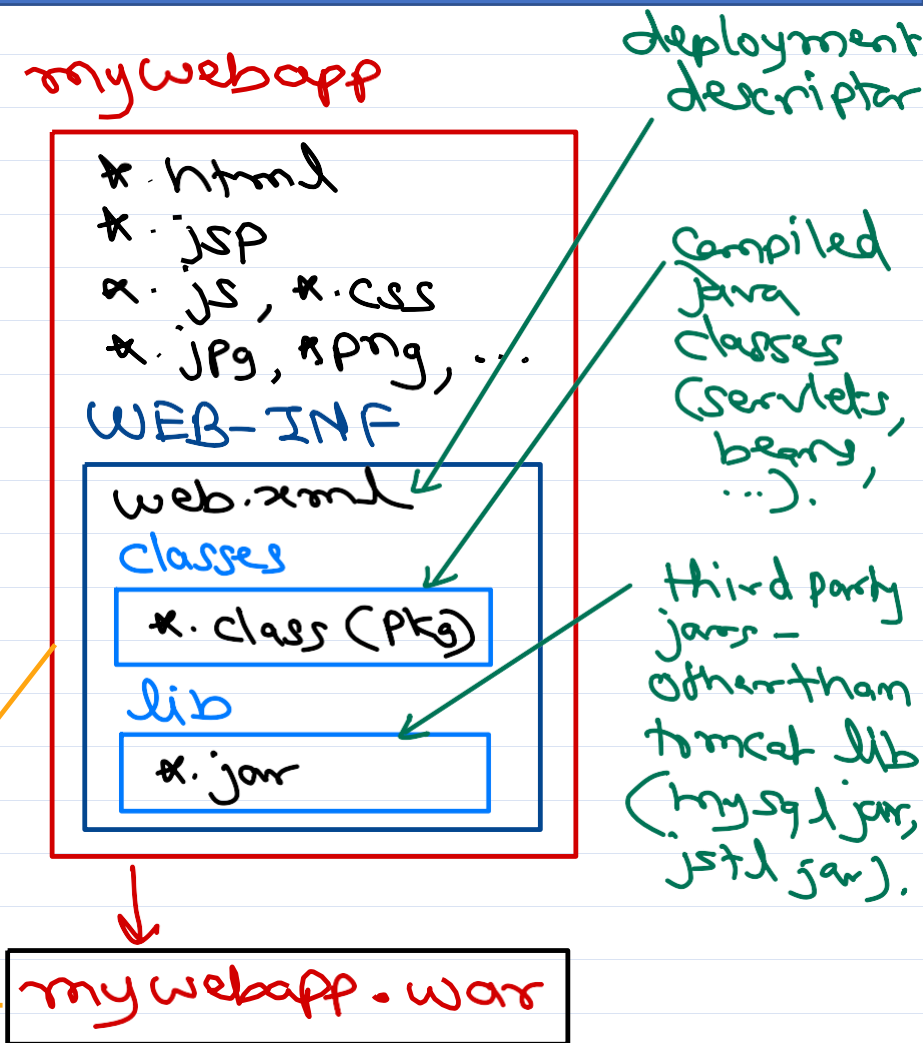
Java Web Application

* To start/test tomcat on your system:

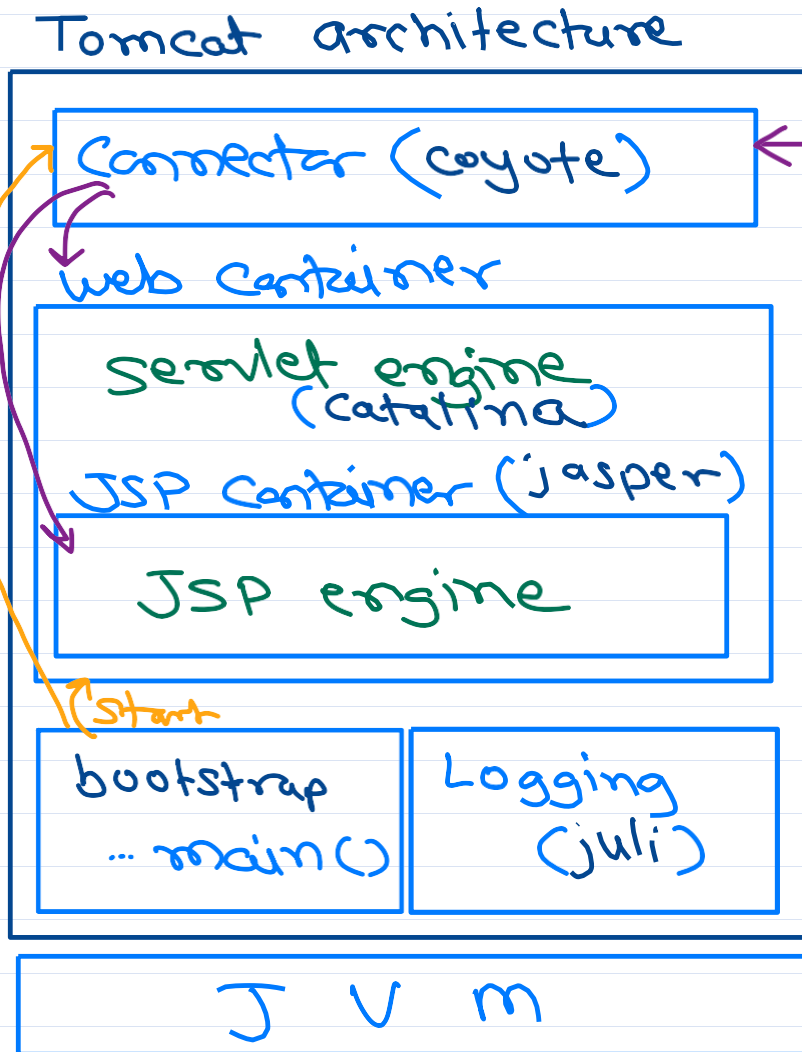
- ① download & extract tomcat.
- ② environment var → JAVA_HOME → path of jdk.
- ③ tomcat dir → bin → startup.bat ✓
- ④ browser → http://localhost:8080
- ⑤ tomcat dir → bin → shutdown.bat ✗

* Tomcat directories

- ① bin: start/stop scripts, tomcat main executable.
- ② Conf: config files of tomcat
e.g. server.xml to set tomcat port (default: 8080)
- ③ lib: tomcat libs/jars (which implements java ee specs) e.g. servlet-api.jar, jsp-api.jar, ...
- ④ logs: web server execution logs (exceptions, ...).
- ⑤ temp & work: temp files created at runtime for appn execution.
- ⑥ webapps: Hot deployment directory.
 - apps to be deployed are copied here.
 - hot ⇒ apps copied while server is on are auto-deployed.



Tomcat architecture



Key parts:

- ① Connector (accept reqs)
- ② web container (req processing)
- ③ web appls running in tomcat.

Request Processing

- ① accept request
- ② url mapping
↳ @webServlet or web.xml
- ③ Servlet object
↳ created on 1st req
↳ accessed from pool on next req
- ④ Response generation
↳ doXYZ() method
- ⑤ HTTP response

Thread model

- ✓ tomcat creates a collection of threads during startup - executor thread pool.
- ✓ on each req to a Connector, a thread from pool is assigned to process that request.
- ✓ upon completion, thread returns back to pool.
- ✓ max num of threads in pool can be configured in server.xml.

